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Set 18: Product Inventory Management

Data Visualization using R

Aim

To analyse product inventory using a bar chart, a stacked bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the inventory data as a data frame. The data has no category column, so categories are assumed.
3. Question 1: draw the bar chart of quantity with `geom_col()`.
4. Question 2: draw the stacked bar chart of quantity by category with `geom_col()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset (Category is assumed, as the data has no category column)
inv <- data.frame(Product_ID = 1:3, Product = c("Product A", "Product B", "Product C"),
                  Quantity = c(250, 175, 300), Price = c(20, 15, 18),
                  Category = c("Category 1", "Category 2", "Category 1"))
print(inv)
print(aggregate(Quantity ~ Category, inv, sum))

# 1. Bar chart - quantity of each product
p1 <- ggplot(inv, aes(Product, Quantity)) + geom_col(fill = "steelblue") +
  geom_text(aes(label = Quantity), vjust = -0.5) +
  labs(title = "Quantity of Each Product", x = "Product", y = "Quantity Available")

# 2. Stacked bar chart - products within categories
p2 <- ggplot(inv, aes(Category, Quantity, fill = Product)) + geom_col() +
  labs(title = "Quantity by Category", x = "Category", y = "Quantity")

# 3. Table
tb <- tableGrob(inv[, 1:4], rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Inventory Dashboard")

ggsave("set18_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set18_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set18_q3.png", arrangeGrob(tb, top = "Product Inventory Table"), width = 5, height = 2, bg = "white")
ggsave("set18_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

	Product_ID	Product	Quantity	Price	Category
1	1	Product A	250	20	Category 1
2	2	Product B	175	15	Category 2

3	3 Product C	300	18 Category 1
	Category Quantity		
1	Category 1	550	
2	Category 2	175	

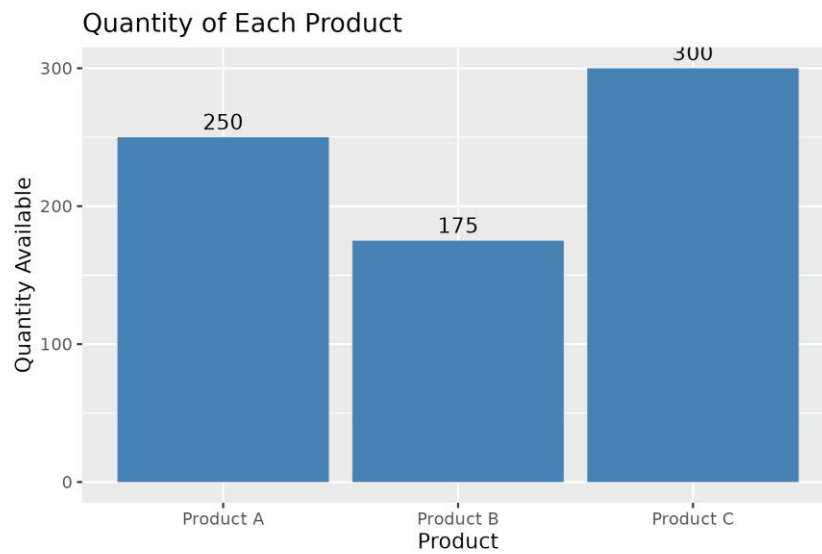


Figure 18.1: Q1 - Quantity of each product

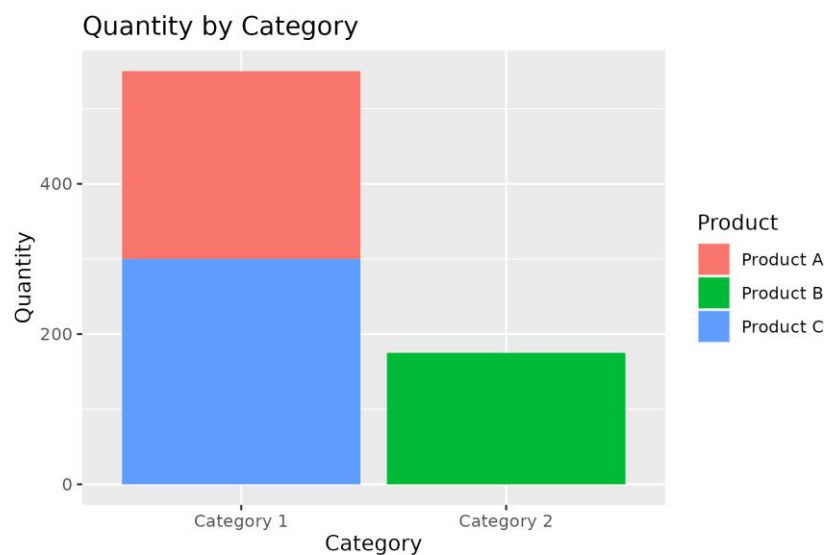
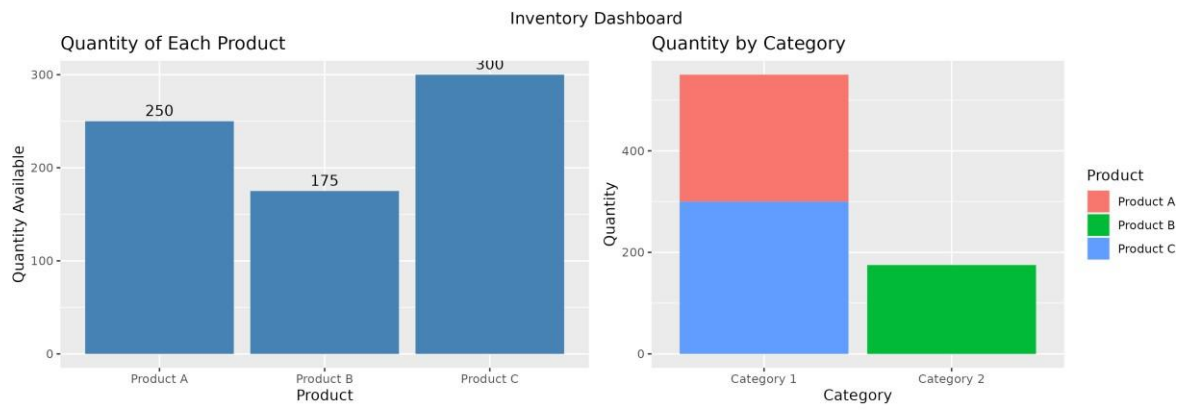


Figure 18.2: Q2 - Stacked bar chart by category (assumed)

Product Inventory Table

Product_ID	Product	Quantity	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Figure 18.3: Q3 - Inventory table



Product_ID	Product	Quantity	Price
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

Figure 18.4: Q4 - Dashboard

Result

- Product C has the most stock (300) and Product B the least (175).
- Total stock is 725 units.
- Categories are assumed, so the stacked chart is for demonstration.

